Evaluating Temporal Coordination Detection: False-Positive Control and Robustness under Synthetic Engagement Behaviour

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Detector V1 (MIT licence): https://github.com/heyuxi6666/social-coordination-detector

Evaluation code available on request.

Abstract

Coordinated inauthentic engagement is often detected through synchrony, yet legitimate audiences also react together. We evaluate V1, an interpretable non-machine-learning detector that flags accounts only when they repeat a personal, fixed like delay across posts. We compare it with an engagement-overlap baseline and a SynchroTrap-style time-bucket baseline tuned on separate seeds. On 1,290 held-out datasets (43 configurations × 30 seeds) of a synthetic benchmark, V1 flagged none of 390 organic-only datasets. The tuned baseline, by contrast, flagged every dataset containing an audience that reacted with a mean delay of 120 s or less. V1 was clearly worse when coordinated accounts randomised their delays (F1 ≤ 0.04 vs 1.00), at 40% participation (0.00 vs 0.43) and under 40 s of timing jitter (0.42 vs 1.00). The results characterise a trade-off within a synthetic model; they do not measure real-world accuracy.

1. Introduction and research questions

Synchrony is a central signal for coordinated behaviour [1–3], but ordinary audiences are synchronised too. V1 therefore tests a narrower regularity: whether the *same accounts repeat their own fixed delay* from post to post. It reports statistical anomalies, never verdicts about intent. **RQ1:** does timing-based detection outperform overlap-only detection, especially on false positives? **RQ2:** how robust is it to jitter, partial participation, wide or randomised delays and heavy users? **RQ3:** how does account rotation affect it? The contribution is an evaluation, not a new principle. It has three parts: an open fixed-offset permutation test, a reproducible benchmark with disjoint tuning and held-out seeds, and a documented account of where the design helps and where it fails.

2. Related work

CopyCatch [1] detects lockstep Page-Like behaviour on Facebook from the user–Page graph and like timestamps alone. SynchroTrap [2] clusters accounts whose actions on the same object, such as an IP address or a target page, fall within a matching window. It uses Jaccard similarity and connected components, and was deployed in five applications at Facebook and Instagram. Related graph methods find synchronised or dense suspicious structure [3, 4]. Pacheco et al. [5] build coordination networks from shared behavioural traces without supervision, and Giglietto et al. [6] define coordinated link sharing through a short time window. Newer spambots evade older detectors [7, 8], and supervised graph-based detection uses benchmarks such as TwiBot-22 [9].

3. Methods

Table 1. Signals used by each method (✓ = used, – = not used).

Signal	A overlap-only	B time-bucket	C V1
Who liked which post	✓	✓	✓
Like timestamps	–	✓ absolute, bucketed	✓ per-account delay between posts
Post author ids	–	–	✓ same-author yardstick
Account-id numbering (metadata)	–	–	✓ id-adjacency indicator
Like and repost counts	–	–	✓ repost-to-like indicator, if supplied

A. Overlap-only is V1's own group-forming step. It links two posts when their liker-overlap coefficient is at least 0.3 and the Bonferroni-corrected hypergeometric $p \leq 10^{-3}$, then flags accounts that liked at least half of a group's posts.

B. Time-bucket co-action is a simplified SynchroTrap-style baseline, not a reimplementation of [2]. It turns each like into an item (post, ⌊time/w⌋) and links accounts that share at least 3 items with item Jaccard ≥ 0.5. Connected components of at least $n_{\min}$ accounts are flagged (originally w = 300 s, $n_{\min}$ = 5). Unlike SynchroTrap's matching window, fixed buckets can split near-simultaneous likes.

C. V1 forms groups as in A and scores them on six indicators: three overlap-based, one timing (the fixed-offset cohort), one account-metadata (consecutive ids) and one count-based (the repost-to-like ratio). For posts a and b and each shared liker i, let $d_i = t_i(b) - t_i(a)$, and let c_{obs} be the size of the largest set of shared likers whose d_i fit in a 20 s window. Over R = 500 seeded re-pairings of second-post times, $p = (1 + \#\{r : c_r \geq c_{\text{obs}}\}) / (R + 1)$ [10]. A cohort is *fixed-offset* if four conditions hold: $p \leq 0.01$; $c_{\text{obs}} \geq 10$; the window centre is at least 120 s, so it is not a single browsing session; and its like times on a span at least max(60 s, 5 × the range of its d_i), so it is not a single burst.

A group is rated *strong* if at least 3 indicators are measurable and all of them trigger, except single-author groups that lack both timing and same-author support. It is rated *weak* if timing is measurable but finds no fixed-offset cohort, and otherwise *moderate* if at least 2 indicators trigger. Under a rule fixed before the main experiments, strong and moderate groups are flagged, and their fixed-offset cohorts are the flagged accounts. V1's code was not modified.

4. Experimental setup

Synthetic world. Each dataset has 2,000 accounts and 30 posts over 48 h. Posts receive about 40 organic likers (±30%) with exponential delays (mean 2,400 s), and 10% of accounts are heavy users with five times the selection weight. In coordinated datasets, 40 accounts like 6 posts by 3 authors. Each account reuses a personal delay (spread over 120 s) with Gaussian jitter (sd 2 s). By default the generator supplies no repost counts and gives group accounts random ids. We vary jitter, participation, rotation, cohort spread, and a *random-delay* mode that redraws every delay. The negative scenarios are organic-only data; 35 followers liking one author's 3 posts, with a mean delay from 15 s (a burst) to 1,800 s (a loyal audience); and heavy-user weights up to 50.

Splits and tuning. Seeds 2000–2019 were used only to tune B, over 42 cells (w = 60–3,600 s, $n_{\min}$ = 3–20) and 11 scenarios (7 coordinated, 4 negative). Maximising mean F1 minus mean false-positive rate selected **w = 2,400 s, $n_{\min}$ = 15**; its tuning score was 0.691, against 0.677 for the original setting. All results use held-out seeds 5000–5029. A and V1 were not tuned, and the protocol was fixed before tuning.

Metrics. We report mean per-seed account F1, where group detection requires post-set Jaccard ≥ 0.5, and false-positive rates with Wilson intervals [11]. Paired per-seed differences compare coordinated configurations on F1 and organic-only ones on the false-positive indicator; the two are never combined.

Intervals. The pre-registered paired intervals (mean ± 1.96 SE) fell outside [−1, 1] in 4 of 129 comparisons (lowest −1.032). We therefore report seeded percentile-bootstrap intervals (10,000 resamples) [12]; the verdicts are unchanged except in one negligible comparison. All configuration-level wins and losses are exploratory, because no multiple-comparison correction was applied.

5. Results and discussion

RQ1: false positives. V1 flagged **none of 390 organic-only held-out datasets** (Wilson upper bound 0.010). On plain organic and heavy-user data (150 datasets), every method was nearly clean, and A flagged 2. On follower-audience data (240 datasets), A flagged 240, original B 142, tuned B 167, and V1 none. Both B variants flagged every audience with a mean delay of 120 s or less (Fig. 1a). Tuning shifted their false-positive region rather than removing it: tuned B flagged 0.97 of datasets at 300 s and 0.53 at 900 s. All four detectors found the plain coordinated group in all 30 datasets, but A absorbed an adjacent loyal audience (F1 0.68). V1's *strong* tier never fired on default data (0 of 30 datasets), because the repost indicator was unmeasurable and the id-adjacency indicator did not trigger. In a diagnostic scenario that plants consecutive ids and inflated repost counts, which are exactly what those indicators measure, the *strong* tier fired in 30 of 30 diagnostic datasets. The flagging rule therefore includes *moderate*.

Table 2. Default settings, 30 held-out seeds. The first three rows give mean account F1; the last three give the share of organic-only datasets flagged, with Wilson 95% intervals.

Scenario	A overlap	B original	B tuned	C V1
Coordinated group	0.98	1.00	1.00	1.00
Coordinated + loyal audience	0.68	1.00	1.00	1.00
Coordinated, V1 *strong* tier only	–	–	–	0.00
Organic only	0.00 [0.00, 0.11]	0.00 [0.00, 0.11]	0.00 [0.00, 0.11]	0.00 [0.00, 0.11]
Loyal audience (mean delay 1,800 s)	1.00 [0.89, 1.00]	0.00 [0.00, 0.11]	0.03 [0.01, 0.17]	0.00 [0.00, 0.11]
Notification burst (mean delay 45 s)	1.00 [0.89, 1.00]	1.00 [0.89, 1.00]	1.00 [0.89, 1.00]	0.00 [0.00, 0.11]

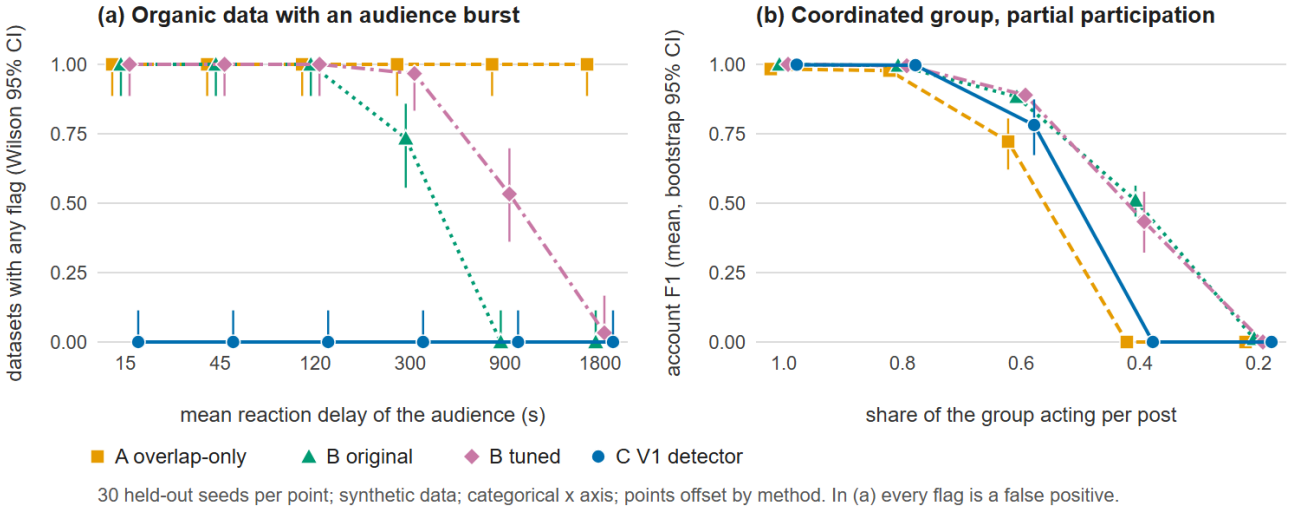


Figure 1. (a) Share of organic-only datasets flagged, by the mean reaction delay of a 35-account audience (Wilson 95% intervals); every flag is a false positive. (b) Mean account F1 under partial participation (percentile-bootstrap 95% intervals). 30 held-out seeds per point.

RQ2: robustness. V1 held at 1.00 up to a 2,000 s cohort spread. At that spread the original B fell to 0.91, while tuned B (0.99) was indistinguishable from V1 (Table 3). V1 lost clearly in three conditions. With randomised delays, F1 $\leq$ 0.04. This follows from its design; A and tuned B still found the overlapping group. With participation at 0.4, the overlap step V1 shares with A no longer forms a group (Fig. 1b). Under jitter, V1 scored F1 0.99 at 20 s, then 0.42 at 40 s and 0.13 at 80 s, reflecting its fixed 20 s window. Heavy users reduced A's F1 to 0.85 at weight 50 but barely affected B or V1.

Table 3. Mean account F1 (30 held-out seeds) and the paired difference V1 − tuned B, with a percentile-bootstrap 95% interval. The standard errors of all F1 means shown are at most 0.07.

Setting	A	B original	B tuned	C V1	V1 − B tuned
Cohort spread 2,000 s	0.98	0.91	0.99	1.00	+0.01 [−0.00, +0.02]
Random delays, range 120 s	0.98	1.00	1.00	0.04	−0.96 [−1.00, −0.90]
Random delays, range 2,000 s	0.98	0.01	1.00	0.00	−1.00 [−1.00, −1.00]
Participation 0.6	0.72	0.88	0.89	0.78	−0.11 [−0.21, −0.02]
Participation 0.4	0.00	0.51	0.43	0.00	−0.43 [−0.54, −0.32]
Jitter sd 40 s	0.98	1.00	1.00	0.42	−0.58 [−0.71, −0.46]
Jitter sd 80 s	0.98	1.00	1.00	0.13	−0.87 [−0.95, −0.78]
Rotation 0.1	0.82	0.83	0.83	0.91	+0.08 [+0.07, +0.09]
Rotation 0.4	0.52	0.41	0.42	0.30	−0.11 [−0.22, −0.01]
Rotation 0.6	0.06	0.18	0.16	0.06	−0.10 [−0.15, −0.04]

RQ3: rotation. V1 led only at rotation 0.1 (0.91 vs 0.83). A possible but untested explanation is that V1 combines cohorts from individual post pairs, whereas B needs three shared items. From rotation 0.4 V1 trailed B and returned fragments of the group (0.57 per dataset, against 0 for B). Its tracer recovered 87% of coordinated posts at rotation 0.2, 33% at 0.4 and 0% at 0.8.

Exploratory summary. Against tuned B, V1 was better in 1, worse in 12 and not distinguishable in 17 of the 30 coordinated configurations (compared on F1). On the 13 organic-only configurations (compared on false

positives) it had fewer false positives in 6, more in 0 and no clear difference in 7. Against A, the counts were 20 / 6 / 4 and 8 / 0 / 5. At default settings the median runtime per dataset was 176.5 ms for V1, 3.0 ms for tuned B and 0.7 ms for A, on one machine.

6. Limitations and future work

The generator and V1 share an author and a model of coordination, so V1 is tested on the behaviour it was designed for. Nothing here estimates real-world precision or recall, and a flag is a statistical anomaly, not evidence of intent or of who controls the accounts. The synthetic world lacks follower graphs, daily cycles and adaptive adversaries; jitter does not grow with the gap between posts; and factors were varied one at a time at one dataset size. Only B was tuned, on scenario families shared with the test set (though on different seeds), and bootstrap intervals from 30 seeds can under-cover slightly. Future work includes real-data validation, adaptive adversaries, false-discovery-rate control [13], and a two-stage design in which B proposes groups and V1 filters them.

7. Conclusion

In this synthetic model, requiring repeated per-account delays bought near-complete rejection of legitimate synchronised audiences: V1 flagged none of 390 organic-only datasets. The price was lower recall when coordinated accounts randomised their delays, acted on only part of the posts, jittered by 40 s or more, or rotated. V1 is best viewed as an interpretable, low-false-positive *filter*, not a better detector overall, and its behaviour on real platform data is untested.

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